



STUDENT INNOVATION CHALLENGE Thailand 2024

ENTRY FORM

ORGANIZED BY:



Entry Number:

(Official use only)

1. Title of the Project / Device

ALL Wheelchair: AI Motion Tracking System for Monitoring Health and Activity

2. Particulars of Applicant

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2.1. Particulars of other team members

****Please enter name(s) in full [no nicknames as name(s) will be use for certificate and documentation].***

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3. Categories

Please tick on one of the following

☐ Design Category

☒ Technology Category

4. Project Description

Give a brief description of your project: Objectives of the project. Who are the target users? Why does the user need this? Have the users been consulted? Feature design or technologies used, safety precaution, etc. You may also include some pictures or illustrations of your project.

(*Please try not to exceed 1000 words)

Nowadays, there are many people who use wheelchairs. Because wheelchair is a device that helps facilitate mobility for users with various mobility limitations, such as those who are with disabilities and the frail elderly. Moreover, the global elderly population (aged 65 years and older) is approximately 9% of the total population. This percentage equates to 703 million elderly worldwide. The number is expected to increase significantly in the coming decades due to aging populations in many countries. Therefore, the ratio of the frail elderly who need to use wheelchairs will tend to increase as well. Not only the frail elderly but also disabled person who must need to use wheelchairs. According to the World Health Organization (WHO), there are more than 65 million wheelchair users worldwide which is a huge number.

In fact, a wheelchair user is twice the risk of developing health conditions. The developers realize the importance of encouraging wheelchair users to exercise and have more physical activities which are the basic needs of humans. However, wheelchair users may face many struggles in society in their daily life due to limited ability. Our project "ALL Wheelchair" mainly focuses on monitoring health and activities including exercising purposes for wheelchair users. Our project is developed from "YES Wheelchair" project which analyzes the motion of wheelchair user by using M5StickC sensor to detect the motion and sending the data to the monitor using Bluetooth classic. The real-time data displayed on the monitor consists of accelerometer and gyroscope readings in the x, y, z axes.

Consequently, we develop the new version called "ALL Wheelchair". The system can analyze these motion data as a specific smartwatch for wheelchair users and we develop specific games for specific groups using sensors on the wheelchair's wheels and on the user's wrist to monitor activities and to control the games. We changed the sensor from M5StickC to XIAO nRF52840 due to the battery capacity. The

XIAO nRF52840 have got the power 5000mAh which can last 1,000 hrs. with non-stop monitoring then send motion data using Bluetooth Low Energy to the computer and convert it into keyboard input.

The developed ALL Wheelchair system proved to be a significant high reliability and validity system compared to a 2D motion analysis with $P < 0.05$ and acceptable level of percentage error $< 1\%$ in a study conducted on 30 participants who performed 4 skill-related fitness tests for wheelchair users.

Our project was inspired by Nintendo Wii which uses Wii mote to detect the wheelchair and user's motion and change those motion to control in game. Our designed games are realistic which allows users to experience like they play real sport games such as wheelchair tennis game and wheelchair racing game. We also have created our specific games more than 20 games such as Astro Avenger, Ball game, and Flappy bird which have different methods to control the character. Each game aims for different physical measurement outcomes and designed for different target groups such as wheelchair athletes, frail elderly, etc.

It can detect falls of wheelchair users. The system will notify the caregiver when a wheelchair user falls. Moreover, we have developed a Pre-Cautions system to predict the movement by applying the AI to learn the behavior of wheelchair users. So, we can prevent a safer system for them.

Another important feature is its adaptability. We created the system to be a Smart Wheelchair, so we are integrating it with the smart home system. The wheelchair will be able to control various functions and integration with home devices, such as automatically opening doors when a wheelchair user passes by, automatically turning on lights when a wheelchair user enters a room, and automatically even adjusting the suitable temperature from the air conditions especially when the wheelchair user increases heart rate and heat rate while exercising. Therefore, it will be applied as a smart innovation in allowing the wheelchair users to communicate to the smart house itself.

Targets

- Wheelchair users
- Healthcare providers
- Caregivers

User Needs

Wheelchair users are not different from general people. They need to live a healthy and happy life. However, they can't reach their goals due to the obstacles of their limitation. As a result, they often face challenges engaging in regular physical activities. Nowadays, there are lack of suitable tracking and monitoring activity devices. Our system addresses these needs by providing an accessible, engaging, and effective means to persuade them to have more daily activities and

exercises through motion-controlled gaming. It doesn't only promote physical activity but also enhances mental well-being of them.

User Consultation

Extensive consultation with 30 wheelchair users aged between 18 – 30 years old and 15 healthcare providers, and caregivers has been suggested to ensure the system meets the specific needs and preferences of the target users.

Features

- **ALL Wheelchair:** For every wheelchair, everywhere, and display through every mobile phone.
- **Wheelchair motion sensor system:** Measures wheelchair movement data, including acceleration, speed, distance.
- **Physical motion sensor system:** Analyze vital signs such as heart rate, fingertip oxygen concentration, energy consumption for tracking their performances.
- **Creative game media system:** Offers engaging games to encourage physical activity and assess the fitness levels of wheelchair users.
- **Data processing system:** Monitors user progress, evaluates individual potential.
- **Communication with caregivers:** Utilizes a server system to relay information to caregivers.
- **User information display:** Presents data analysis graphs and user access history on a website page through mobile phone.
- **Monitoring and fall detection:** Monitoring capabilities to detect falls. Upon detecting a fall, the system promptly notifies the caregiver, enhancing user safety. Additionally, the system employs AI to predict and prevent potential falls by studying and learning the behavior of wheelchair users, thus creating a safer environment.
- **Smart home integration:** Designed to function as a Smart Wheelchair, integrating seamlessly with smart home systems. It can control various home functions such as automatically opening doors, turning on lights, and adjusting the temperature based on the wheelchair user's presence and activity. This integration not only enhances convenience but also supports independent living

5. Additional Information about the Project/Device

Has your Project/Device been submitted to any other competition or won any award? If yes, please provide details.



Figure1: “YES Wheelchair” Grand Prize Award (Platinum Award) at the 2023 Kaohsiung International Invention & Design EXPO in Taiwan



Figure 2: "YES Wheelchair" Gold Medal at the 2023 Kaohsiung International Invention & Design EXPO in Taiwan



Figure 3: "YES Wheelchair" Successfully participated in the 2023 Kaohsiung International Invention & Design Expo which featured 445 entries from 30 different countries



Figure 4: “YES Wheelchair” Internationally Outstanding Inventor Awards Ceremony from the 2023 Kaohsiung International Invention & Design Expo

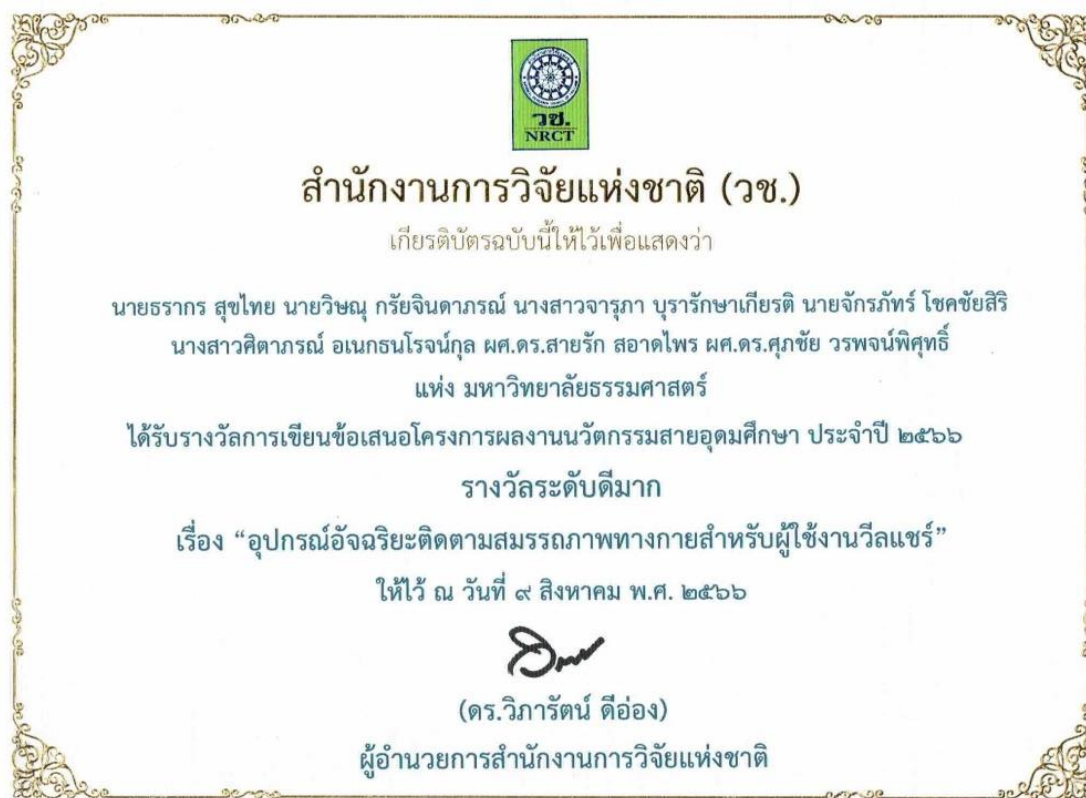


Figure 5: “ALL Wheelchair” Outstanding Proposal Award from National Research Council of Thailand

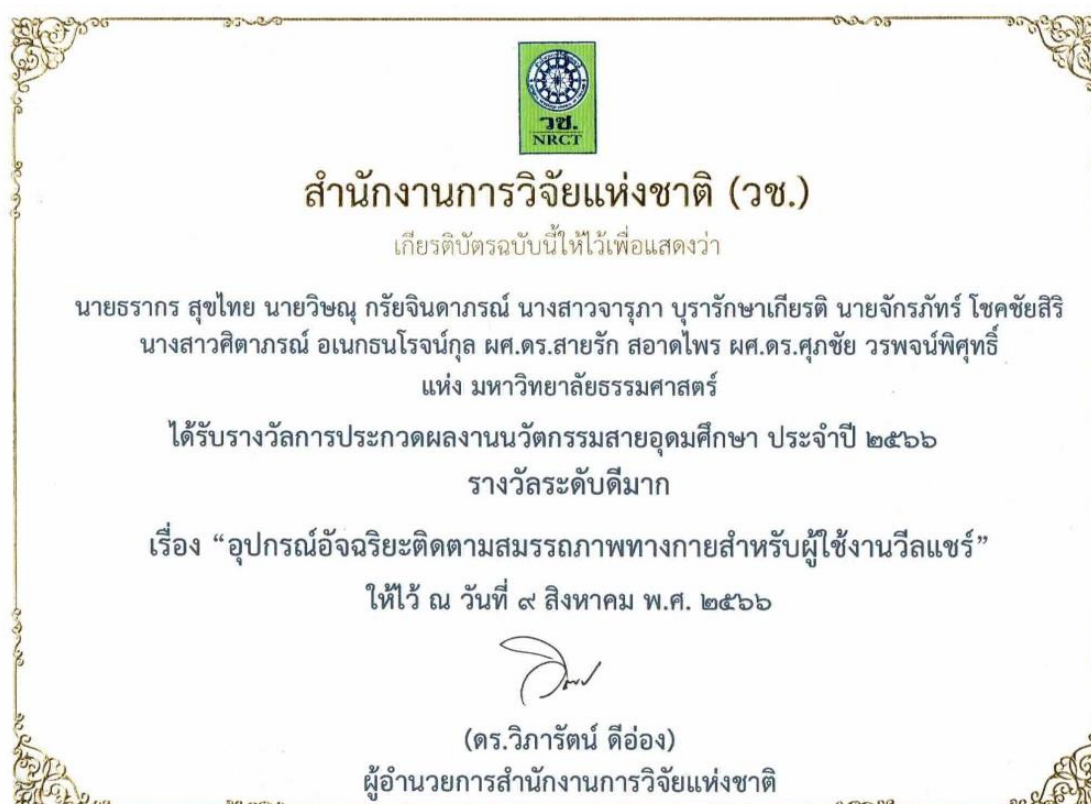


Figure 6: “ALL Wheelchair” Excellent Innovation Award from the Thailand Research Expo 2023



Figure 7: “ALL Wheelchair” Second Runner-up Award from the Sport Science Innovation Contest 2023



Figure 8: Presentation of the invention through YouTube on The Convergence EP.94 under the topic "YES WHEELCHAIR: Innovation for Monitoring the Movement of Wheelchair Athletes"



TSE Research and Innovation ep.04 | YES Wheelchair นวัตกรรมแห่งความร่วมมือระหว่างศาสตร์



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Figure 9: Presentation of the invention through YouTube on TSE Research and Innovation ep.04 under the topic "YES Wheelchair: Innovation of Interdisciplinary Collaboration"

"YES WHEELCHAIR" นวัตกรรมตรวจวัดการเคลื่อนไหว พัฒนา ศักยภาพชีวิตของผู้ใช้งานวีลแชร์

วิทยาศาสตร์ & เทคโนโลยี

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Thai PBS Sci & Tech

แชร์บทความนี้



YES Wheelchair นวัตกรรมตรวจวัดการเคลื่อนไหว พัฒนาศักยภาพชีวิตของผู้ใช้งานวีลแชร์ ให้สามารถติดตามสถานะสมรรถภาพได้เหมือนคนทั่วไป ผลงานโดยอาจารย์และนักศึกษาระบบสารสนเทศฯ วิศวกรรมศาสตร์ ๒๒ ๒๕๖๖

ผศ.ดร.สายรัก สอาดไพร อาจารย์สาขาวิชาวิทยาศาสตร์การกีฬาและการพัฒนากีฬา คณะสหเวชศาสตร์ ม.ธรรมศาสตร์ เล่าถึงจุดเริ่มต้นของการคิดค้น "YES WHEELCHAIR" ว่า ในวงการกีฬาของไทย จะเห็นได้ว่านักกีฬานักกีฬาคนพิการยังขาดตัวช่วยเพิ่มสมรรถภาพร่างกาย จึงได้ร่วมมือกับคณะวิศวกรรมศาสตร์ เพื่อพัฒนานวัตกรรมชิ้นนี้ขึ้นมา



Figure 10: Publication of the invention on the Thai PBS website under the topic "YES WHEELCHAIR: Innovation for Monitoring Movement to Enhance the Quality of Life for Wheelchair Users"

'YES WHEELCHAIR' – THAMMASAT STUDENTS WIN MOTION ANALYSIS INNOVATION FOR WHEELCHAIR ATHLETES.

Congratulations to the Thammasat student team for winning the prestigious international award for their project, YES Wheelchair, an innovative motion detection system for wheelchair users.

Wednesday 10 January 2024



The student team from Thammasat University has been awarded the highest honor, the Platinum award, and a gold medal at the 2023 Kaohsiung International Invention & Design Expo in Taiwan for their innovation project titled **"YES Wheelchair: motion analysis system for wheelchair users."**

YES Wheelchair innovation is a collaboration between students and professors from the Faculty of Allied Health Science and the Faculty of Engineering, Thammasat University. The students from the Faculty of Allied Health Science include Wisanu Kraijindaporn and Tharakorn Sukthai. The students from the Faculty of Engineering include Sitaphon Anekthanarojkul, Jakphat Chokchaisiri, Nattawat Nilpinij, and Chutipon Treerattananurak. The project is supervised by Asst. Prof. Dr. Sairag Sa-adprai from the Department of Sports Science and Sports Development, Faculty of Allied Health Science, and Asst. Prof. Dr. Supachai Vorapojpisut from the Department of Electrical and Computer Engineering, Faculty of Engineering.



Figure 11: Publication of the invention on the Thammasat University website under the topic "YES WHEELCHAIR: Innovation for Monitoring the Movement of Wheelchair Athletes. Thammasat University Faculty and Student Work Wins International Grand Prize

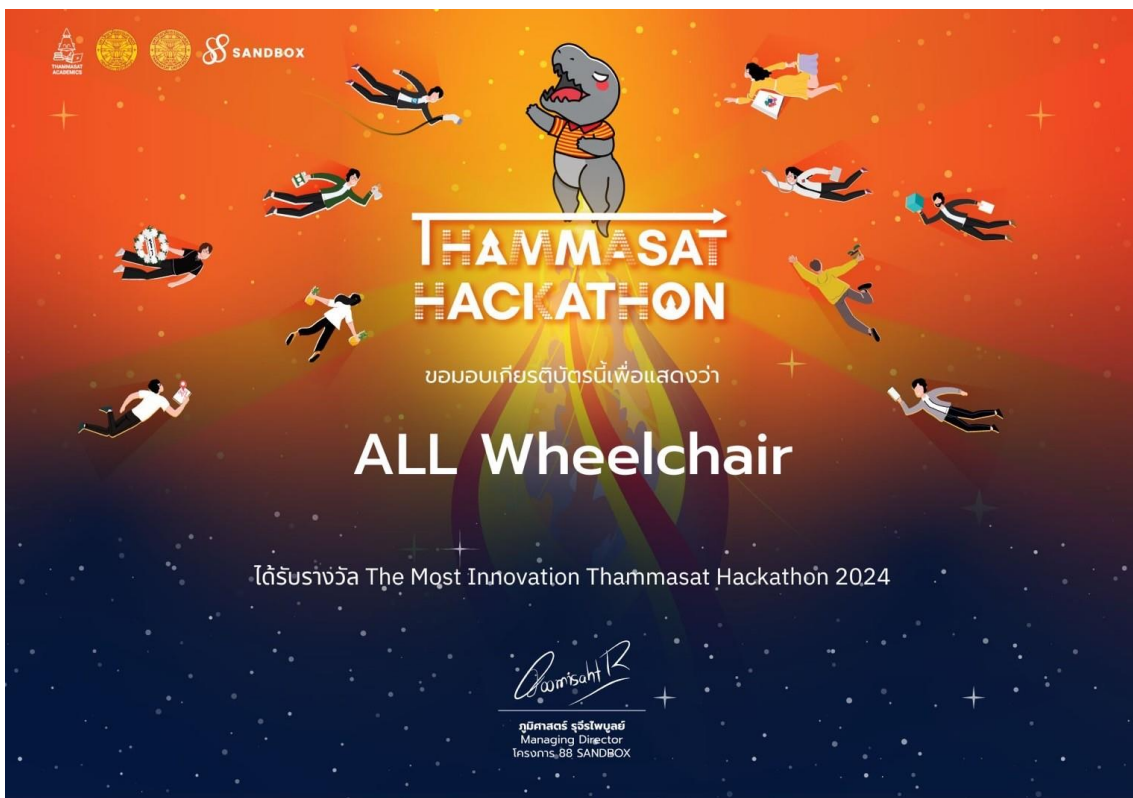


Figure 12: “ALL Wheelchair” The Most Innovation award from The Thammasat Hackathon 2024



Figure 13: Presentation of the invention at 11th Social Business Academia Conference with Nobel Laureate (2006) Professor Muhammad Yunus under the topic "ALL Wheelchair: AI Motion Tracking System for Monitoring Health and Activity"



Figure 14: "ALL Wheelchair" Platinum award from Thailand GEAR TALENT 2024

Other points or aspects that you would like to highlight, if any

System Architecture Overview and Description

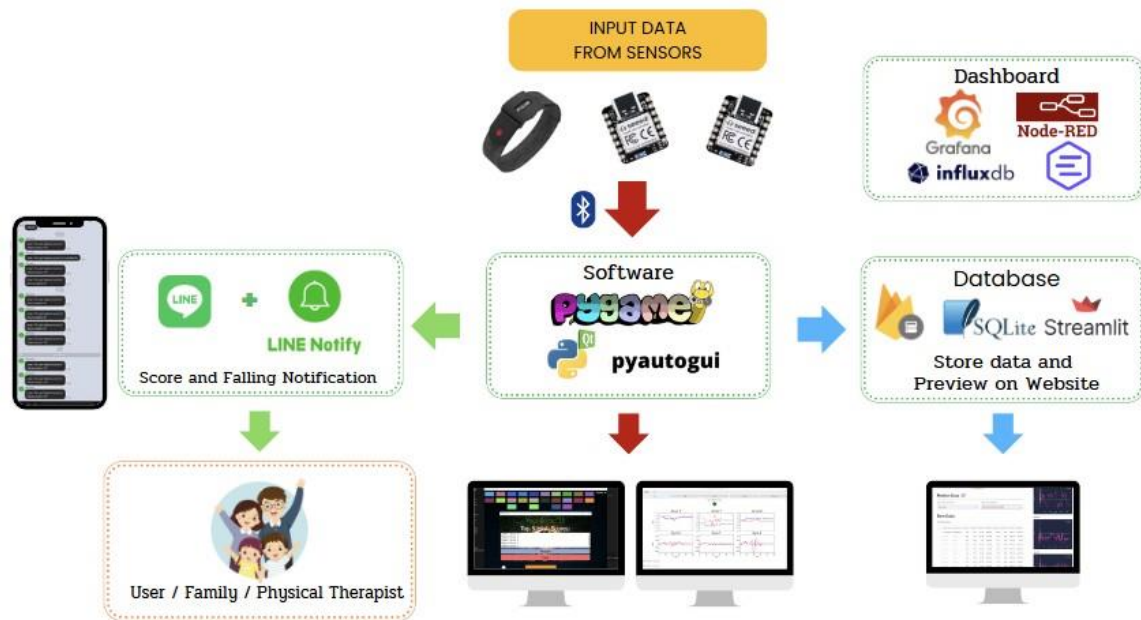


Figure 15: shows “ALL Wheelchair” System Architecture

The system integrates input data from embedded sensors (accelerometers, gyroscopes) on the wheelchair and wristband, transmitting wirelessly via Bluetooth to a central processing unit. Software components include Pygame for motion-controlled gaming and PyAutoGUI for streamlined user interactions. Data management is facilitated through Firebase, SQLite, and Streamlit, with Grafana and Node-RED generating real-time analytics dashboards. InfluxDB manages time-series data storage, complemented by LINE Notify for fall and irregular activity alerts. Mobile notifications via the LINE app ensure continuous user and caregiver updates, enhancing health monitoring and interactive support.

This setup tracks daily activities such as distance, speed, and heart rate, providing a comprehensive view of user mobility. Real-time AI algorithms analyze sensor data for immediate fall detection, triggering timely notifications and preventive actions. The system also integrates seamlessly with gaming consoles, offering enhanced gaming experiences and potential integration with smart home systems for environmental control. Rigorous data security measures safeguard user information during all system interactions.

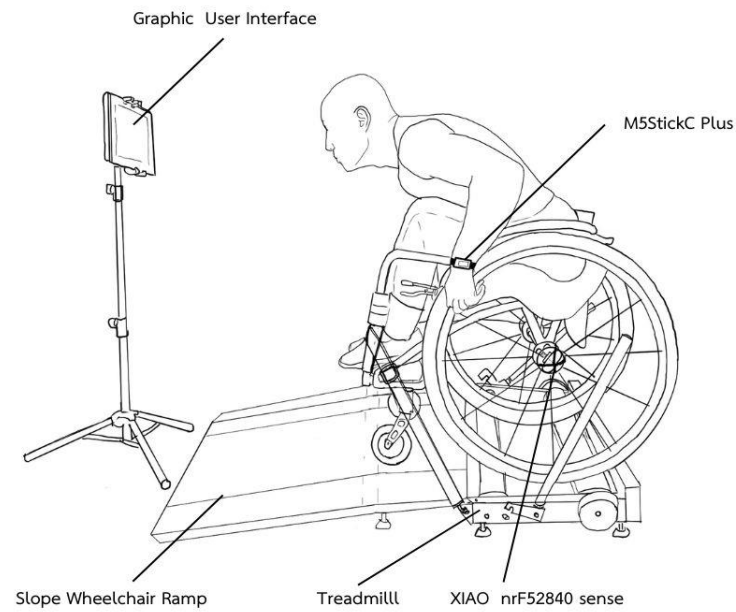


Figure 16: Illustration of the game creation system designed to promote exercise, along with sensor system for monitoring movement in wheelchair users



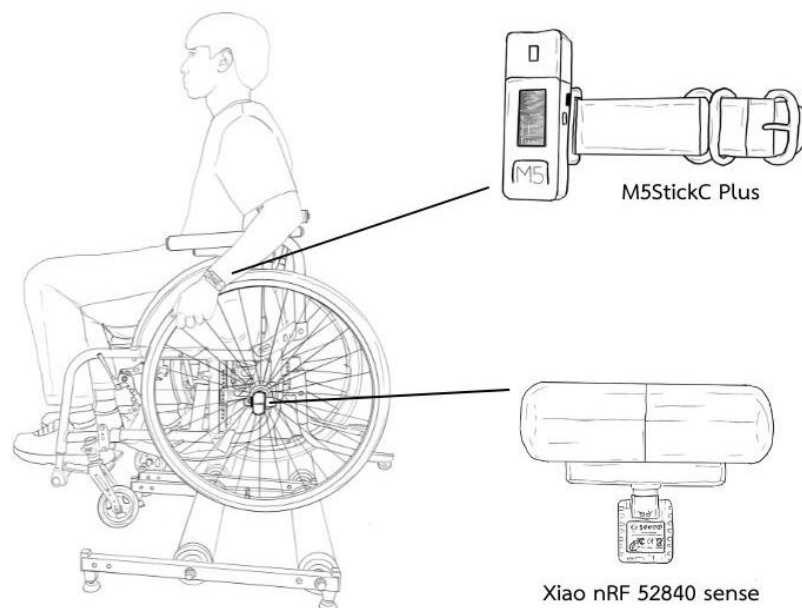


Figure 17: shows the sensor system for monitoring movement in wheelchair users, installed on both sides of the wheel rims, as well as a sensor for analyzing upper limb and basic vital signs at the user's wrist



Figure 18: shows the simple and convenient installation of sensors on wheelchair wheels compatible with all wheelchair models

ALL Wheelchair Core Methodologies and Development Highlights

1. Sensor Integration and Real-time Processing:

- **Sensor Integration:** We integrate high-precision accelerometers and gyroscopes directly into wheelchairs and wristbands. These

sensors continuously capture motion data, which is crucial for tracking user activities.

- Real-time Data Processing: Utilizing Bluetooth connectivity, sensor data is transmitted wirelessly to a central processing unit. This ensures immediate analysis and feedback, enhancing responsiveness and user interaction.
- The developed ALL Wheelchair system proved to be a significant high reliability and validity system compared to a 2D motion analysis with $P < 0.05$ and acceptable level of percentage error $< 1\%$ in a study conducted on 30 participants who performed 4 skill-related fitness tests for wheelchair users.

2. User Interface and Interaction Design:

- Pygame Integration: Our system employs Pygame to create interactive motion-controlled gaming interfaces. This enhances user engagement and provides a dynamic way for users to interact with the system.
- PyAutoGUI for Intuitive Controls: PyAutoGUI facilitates intuitive controls, allowing users to navigate through various functionalities seamlessly. This user-friendly approach ensures accessibility for individuals with varying levels of physical dexterity.

3. Data Management and Visualization:

- Firebase Cloud Integration: We leverage Firebase for cloud-based storage, enabling seamless accessibility to stored data from any connected device. This supports synchronization across platforms and facilitates real-time updates.
- SQLite for Local Data Storage: SQLite manages local databases, ensuring data integrity and enabling offline functionality. This dual approach guarantees that data remains accessible even in environments with limited or no internet connectivity.
- Streamlit for Interactive Data Analysis: Streamlit serves as our web application framework for data visualization and analysis. It provides interactive dashboards that allow users to monitor their health metrics and track progress over time effectively.

4. Analytics, Monitoring, and Security:

- Real-time Analytics with Grafana: Grafana provides robust analytics dashboards for visualizing real-time data trends. This capability enables healthcare professionals and caregivers to monitor user activities and health metrics comprehensively.
- Flow-based Programming with Node-RED: Node-RED facilitates seamless data routing and integration across different systems and devices. It optimizes data flow management, enhancing system efficiency and responsiveness.

- **Secure Data Handling:** Implementing stringent data security protocols ensures user privacy throughout all interactions. This includes encryption methods and secure communication channels to protect sensitive health information.

5. Key Features and Development Phases:

- **Real-time Health Monitoring:** Our system tracks essential health metrics such as distance traveled, speed, and heart rate in real-time. This continuous monitoring provides immediate feedback and enables timely interventions when necessary.
- **AI-driven Fall Detection:** Advanced AI algorithms detect falls automatically and trigger instant notifications to caregivers or emergency services. This proactive feature ensures prompt assistance, enhancing user safety and peace of mind.
- **Integration Possibilities:** Future plans include integrating our system with gaming consoles for interactive gaming experiences and with smart home systems for environmental control. These integrations aim to further empower users and enhance their independence.
- **Comprehensive Testing and Deployment:** Rigorous testing phases ensure that our system meets high standards of functionality, reliability, and usability. User training programs are also implemented to ensure users and caregivers can utilize the system effectively.

6. Future Directions and Continuous Improvement:

- **Expansion into Healthcare Facilities:** We aim to expand our system's deployment into healthcare facilities, enhancing accessibility to advanced health monitoring tools for a broader user base.
- **Iterative Development:** Continuous improvement is central to our approach, driven by user feedback and advancements in technology. This iterative process ensures that our system remains at the forefront of assistive technology, meeting evolving user needs effectively.
- **Monitoring and Fall Detection:** The ALL-Wheelchair system incorporates robust monitoring capabilities to ensure user safety, particularly through advanced fall detection mechanisms. Using high-precision sensors and AI algorithms, the system continuously monitors the user's movements and posture. In the event of a fall, the system detects this automatically and triggers immediate notifications to caregivers or emergency contacts. This proactive approach enhances user safety by ensuring prompt assistance when needed. Moreover, the AI capabilities of the system enable it to learn and predict potential fall risks based on user behavior.

patterns. By analyzing data trends, the system can preemptively alert caregivers to take preventive measures, thus creating a safer environment for wheelchair users.

- **Smart Home Integration:** The ALL-Wheelchair system is designed to function as a Smart Wheelchair, integrating seamlessly with smart home systems to enhance user autonomy and convenience. Through IoT connectivity, the wheelchair can interact with various smart home devices and systems. For instance:
 1. **Automatic Door Opening:** The wheelchair can automatically open doors equipped with smart locks upon approaching them, providing easier access for users.
 2. **Lighting Control:** It can adjust lighting levels based on user preferences or activity, ensuring optimal visibility and comfort.
 3. **Temperature Adjustment:** Integration with smart thermostats allows the wheelchair to adjust room temperatures based on the user's preferences or detected environmental conditions.

By leveraging these methodologies and development highlights, our project seeks to significantly enhance the quality of life for wheelchair users, providing them with comprehensive health monitoring and interactive support through innovative technological solutions.

This integration not only enhances the daily living experience by reducing physical exertion for users but also supports their independence by enabling greater control over their immediate environment. By leveraging these smart home capabilities, the ALL-Wheelchair system promotes a more inclusive and user-centered approach to assistive technology, aiming to improve overall quality of life for wheelchair users.

In summary, through continuous improvement, expansion into healthcare facilities, and the integration of monitoring, fall detection, and smart home functionalities, the ALL-Wheelchair project aims to redefine assistive technology standards. By providing comprehensive health monitoring, proactive safety features, and enhanced user autonomy, the system seeks to significantly enhance the quality of life for wheelchair users, ensuring they have the support and tools necessary to live more independently and safely.

Examples of Designed wheelchair games

In game development using Python and Pygame, key elements like backgrounds, characters, sounds, and controls are crafted. This process starts with importing image files from a folder, like 'icons,' for displaying characters and backgrounds. Control mechanisms, often tied to keyboard inputs, allow player navigation within screen boundaries. Asynchronous functions manage game displays, reducing complexity

and aiding debugging with better error identification and smaller file sizes compared to synchronous methods.



Figure 19: shows in-use demonstration, a participant engages with the system's game-based activity program

Alien Invasion Game

Game Highlights

- Each level tests eye-hand coordination as players must sync with the increasing game speed to progress. This helps evaluate their average eye-hand coordination level based on their scores.

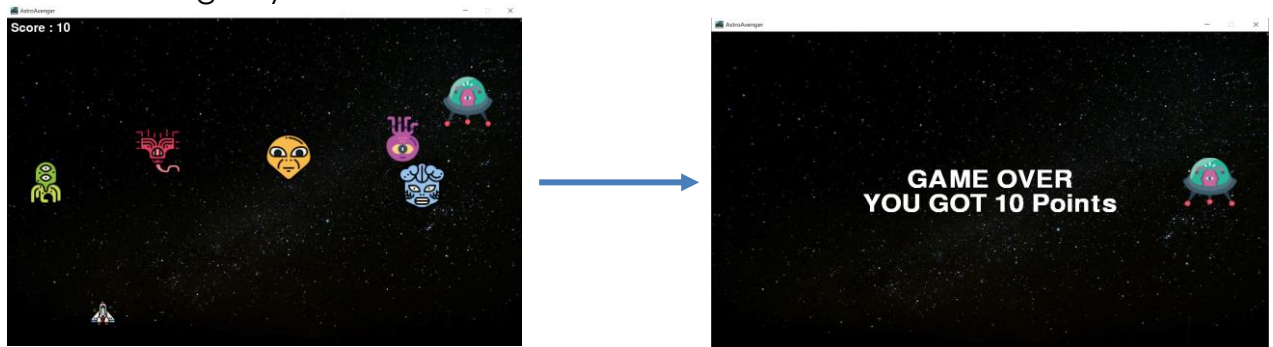


Figure 20: Example gameplay of "Alien Invasion Game"

Bouncing Ball Game

Game Highlights

- "Bouncing Ball" game is an endurance challenge where players must constantly move to score points, with increasing speed adding to the difficulty.

- It's a challenging endurance game where players must constantly move to score points. As the game progresses, the speed increases, adding to the challenge. Towards the end, scoring becomes even tougher, making it a real test of skill.



Figure 21: Example gameplay of "Bouncing Ball Game"

Lucky Bird Game

Game Highlights

- "Lucky Bird" game tests vision and arm muscles, with gameplay falling between fast-paced and slow-paced. It challenges focus, important for elderly ailments involving cognitive decline and muscle weakness.

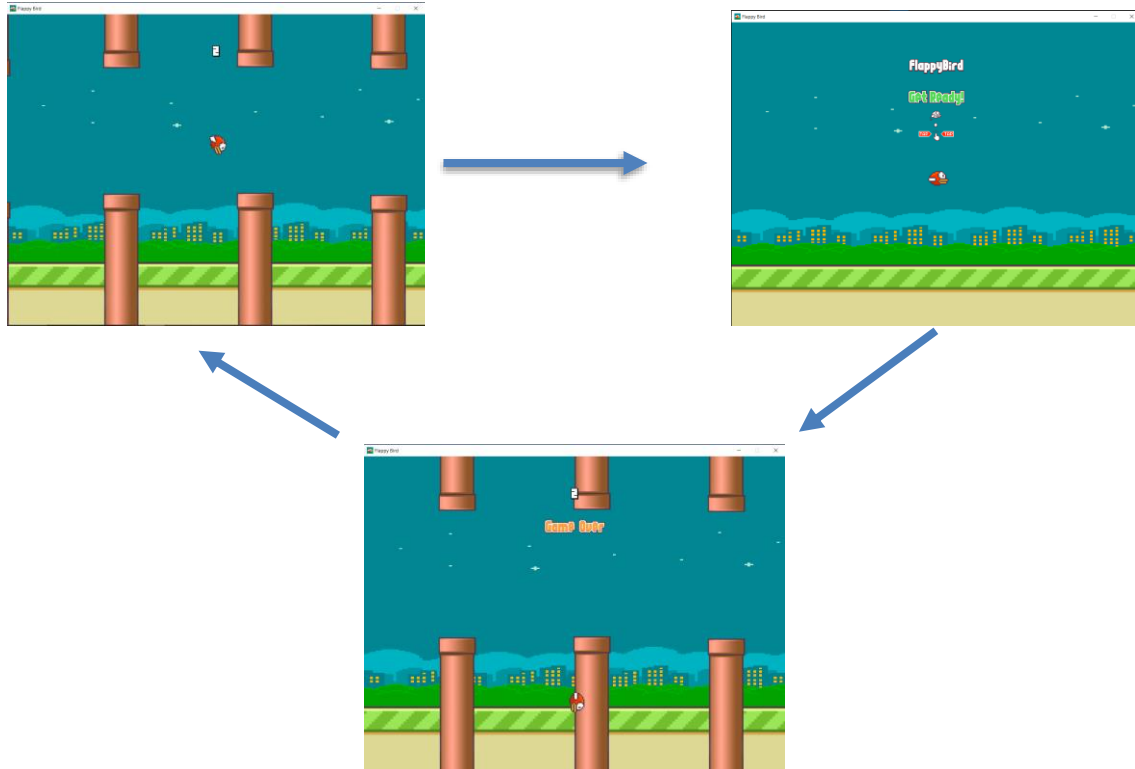


Figure 22: Example gameplay of "Lucky Bird Game"

2D Car Game

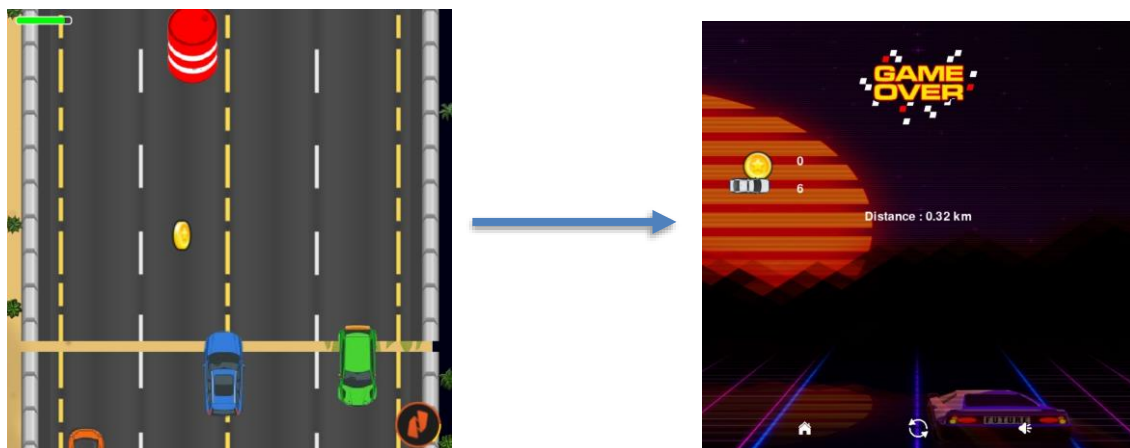


Figure 23: Example gameplay of "2D Car Game"

Aero Blaster Game



Figure 24: Example gameplay of “Aero Blaster Game”

Angry Walls Game

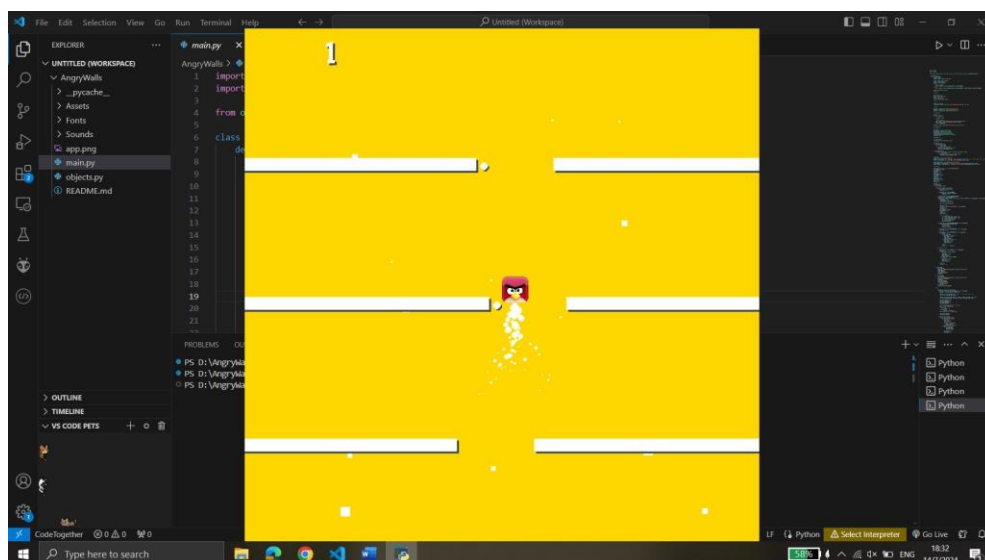


Figure 25: Example gameplay of “Angry Walls Game”

Connected Game

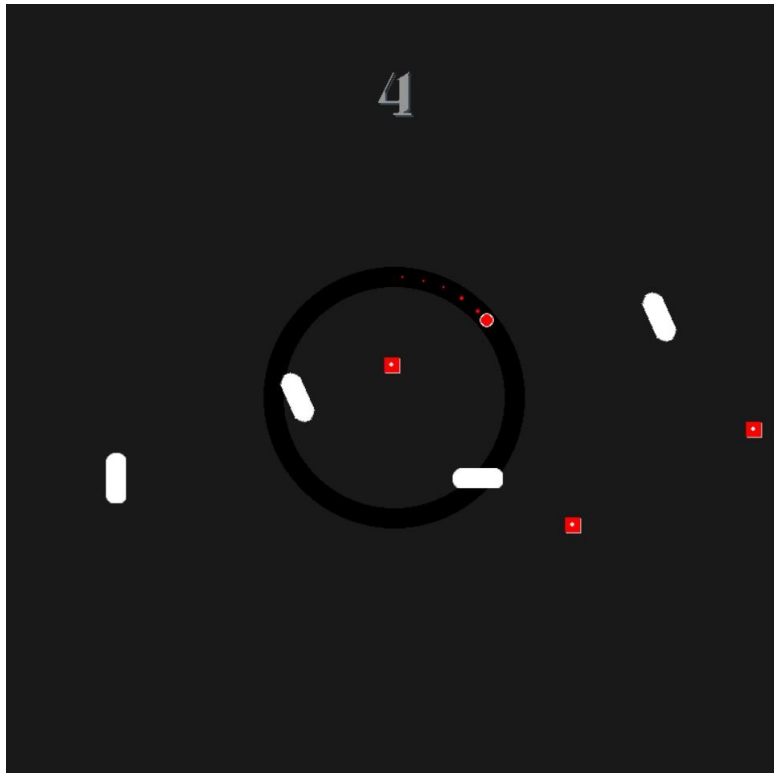


Figure 26: Example gameplay of "Connected Game"

Dino Game

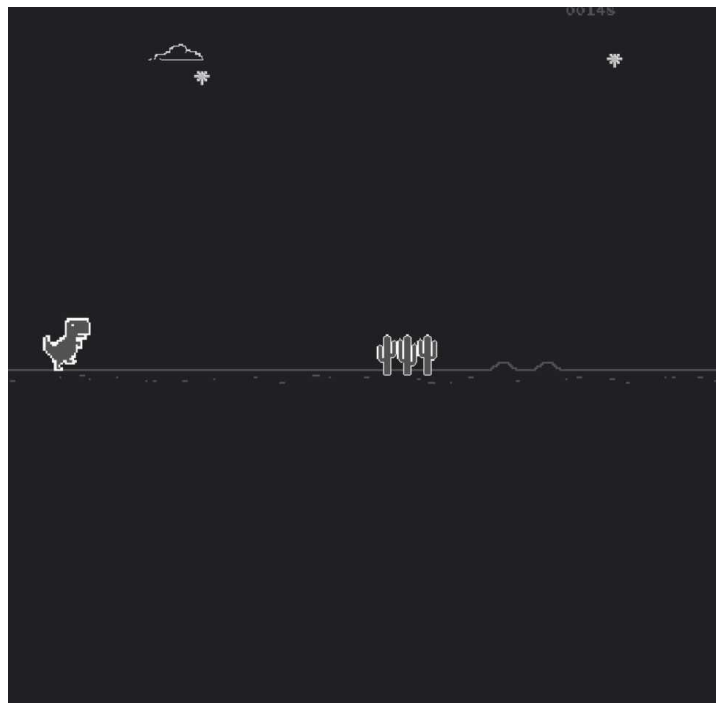


Figure 27: Example gameplay of "Dino Game"

Dodgy Walls Game

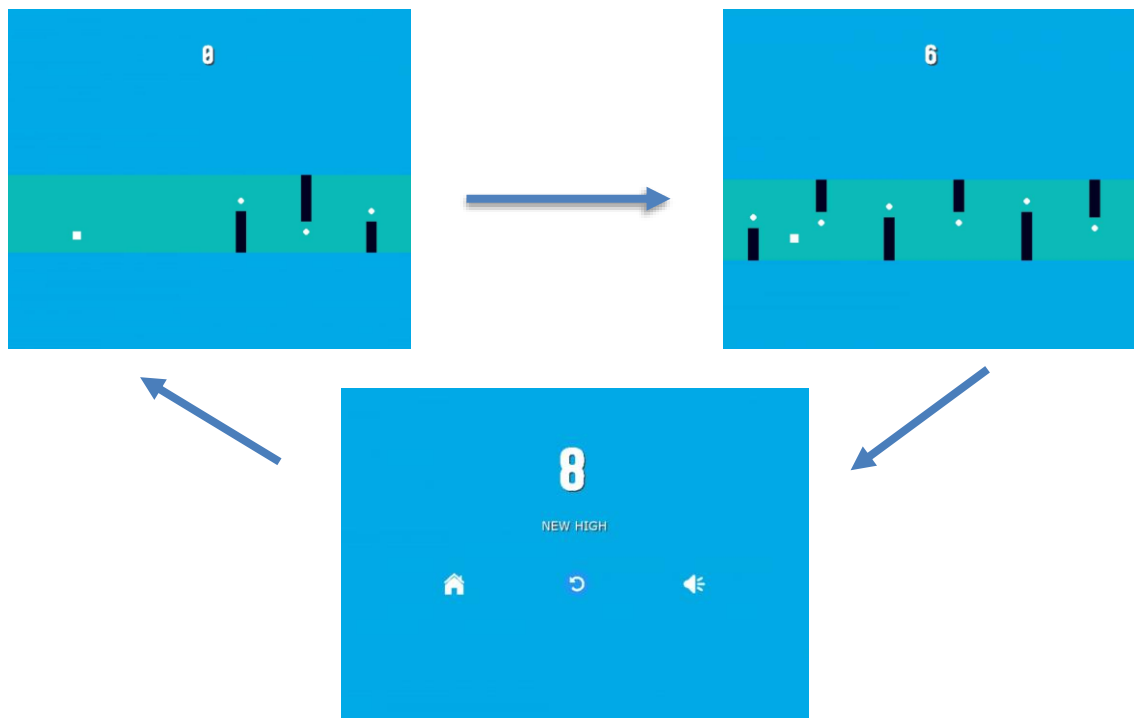


Figure 28: Example gameplay of "Dodgy Walls Game"

Ghost Buster Game

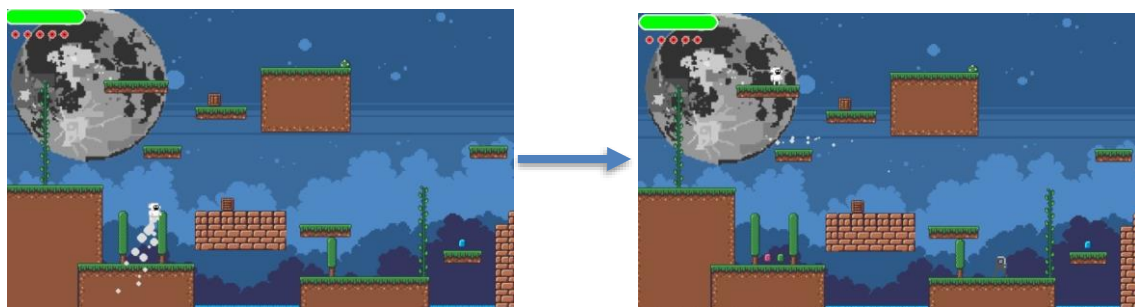


Figure 29: Example gameplay of " Ghost Buster Game"

Hyper Tile Dash Game

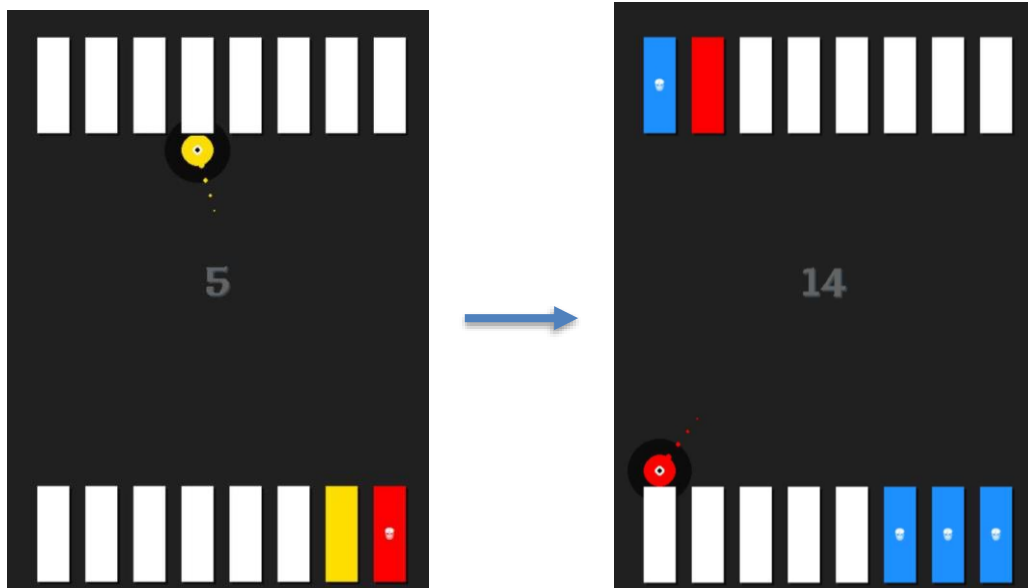


Figure 30: Example gameplay of "Hyper Tile Dash Game"

Jetpack Game

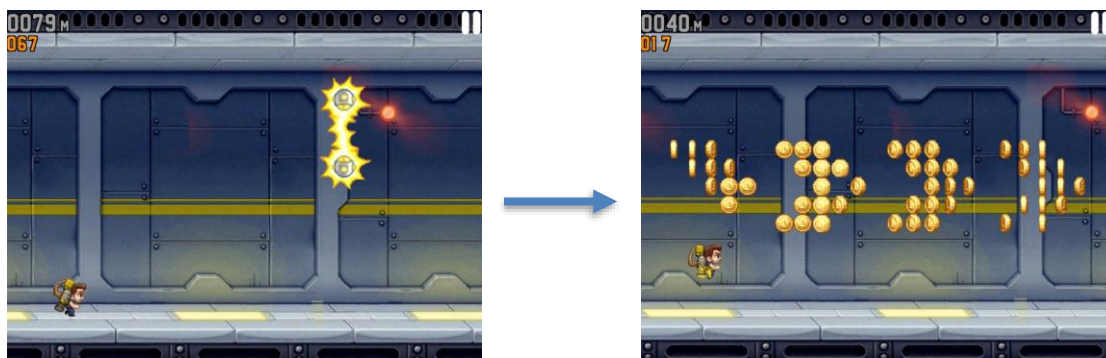


Figure 31: Example gameplay of "Jetpack Game"

Piano Tiles Game

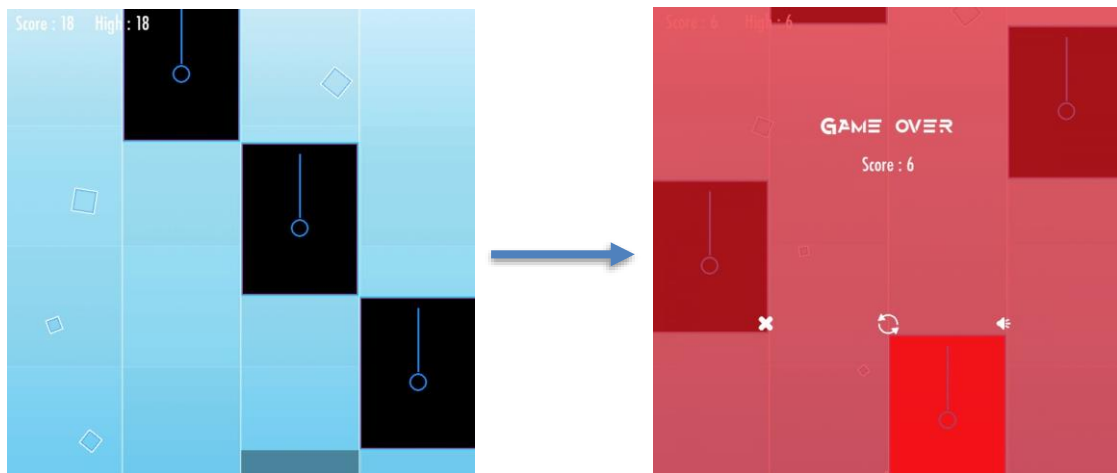


Figure 32: Example gameplay of "Piano Tiles Game"

Table Tennis Game (2 Player Offline)

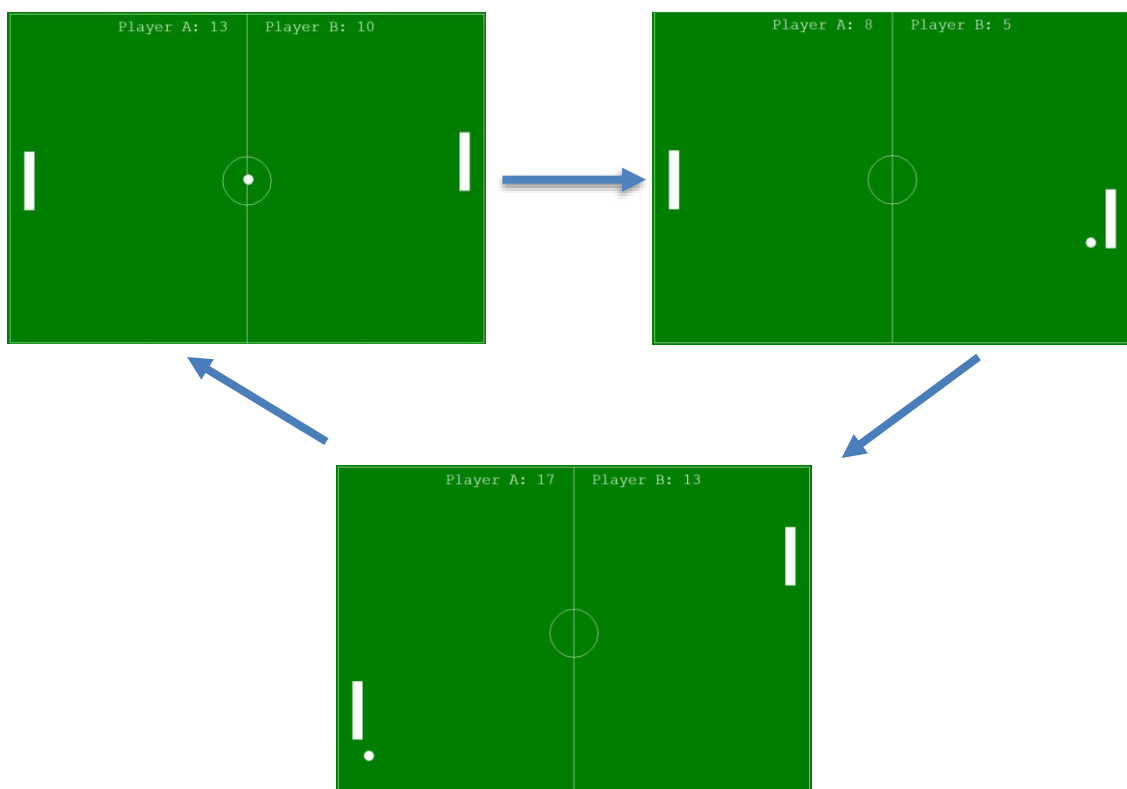


Figure 33: Example gameplay of "Table Tennis Game"

Circle Rush Game

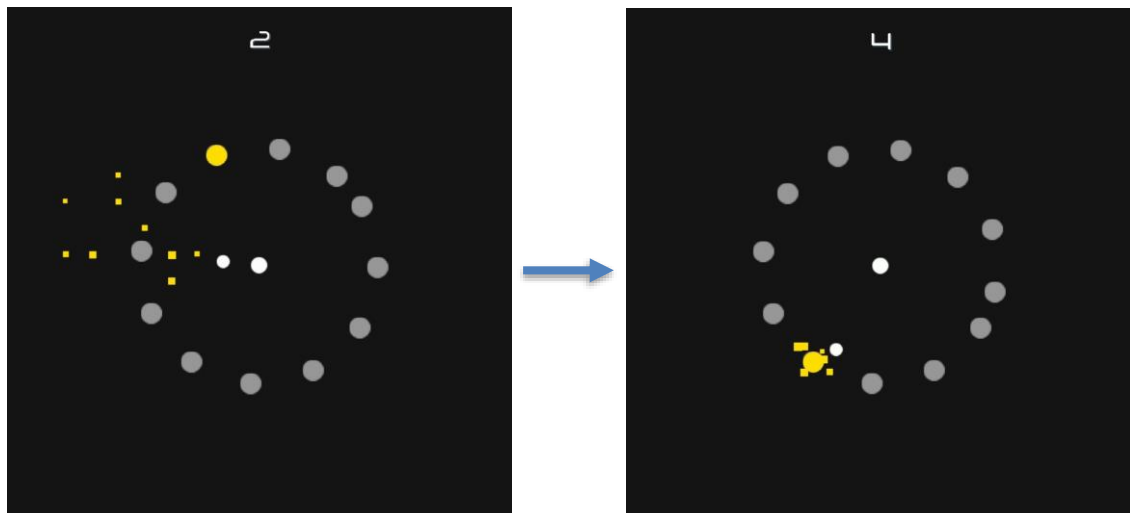


Figure 34: Example gameplay of "Circle Rush Game"

Rotate Dash Game

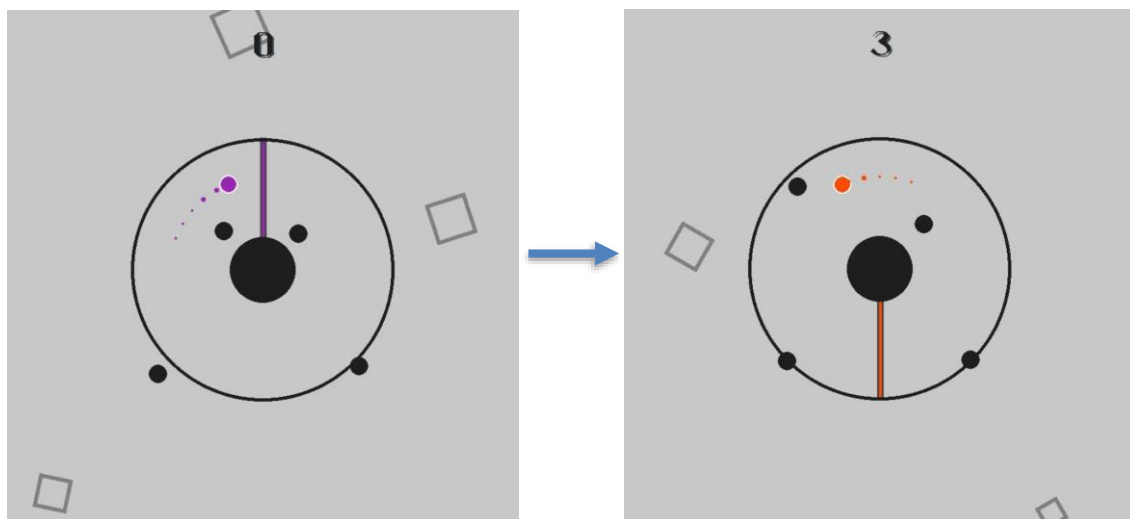


Figure 35: Example gameplay of "Rotate Dash Game"

Tetris Game

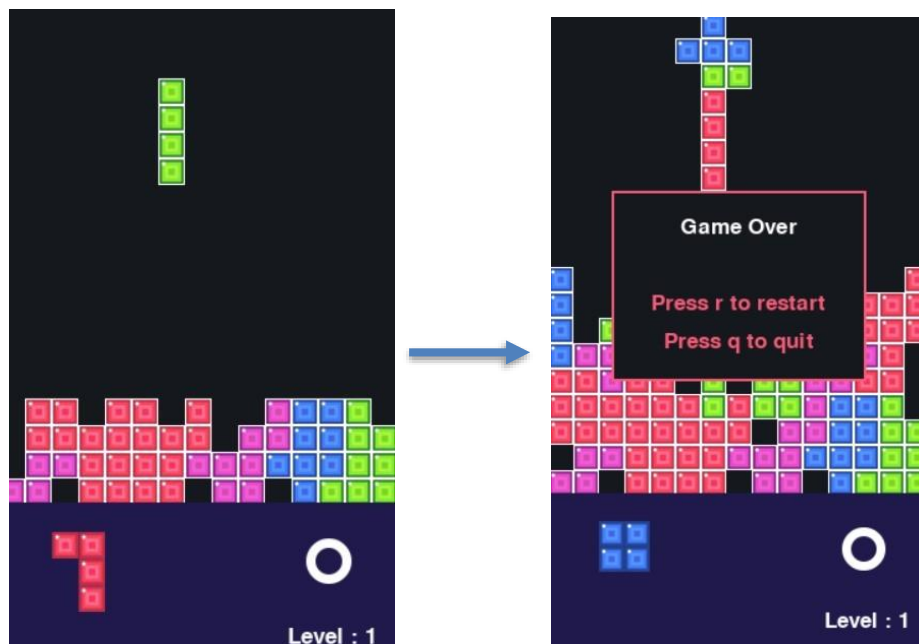


Figure 36: Example gameplay of "Tetris Game"

3D Car Racing Game

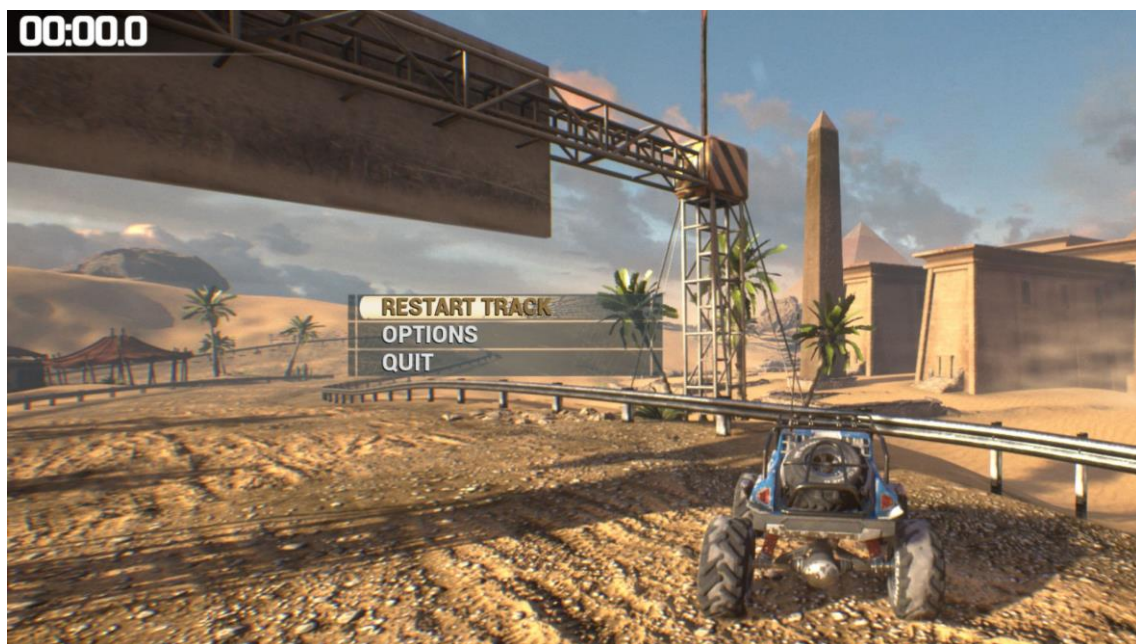


Figure 37: Example gameplay of "3D Car Racing Game"

3D Bomber Man Game

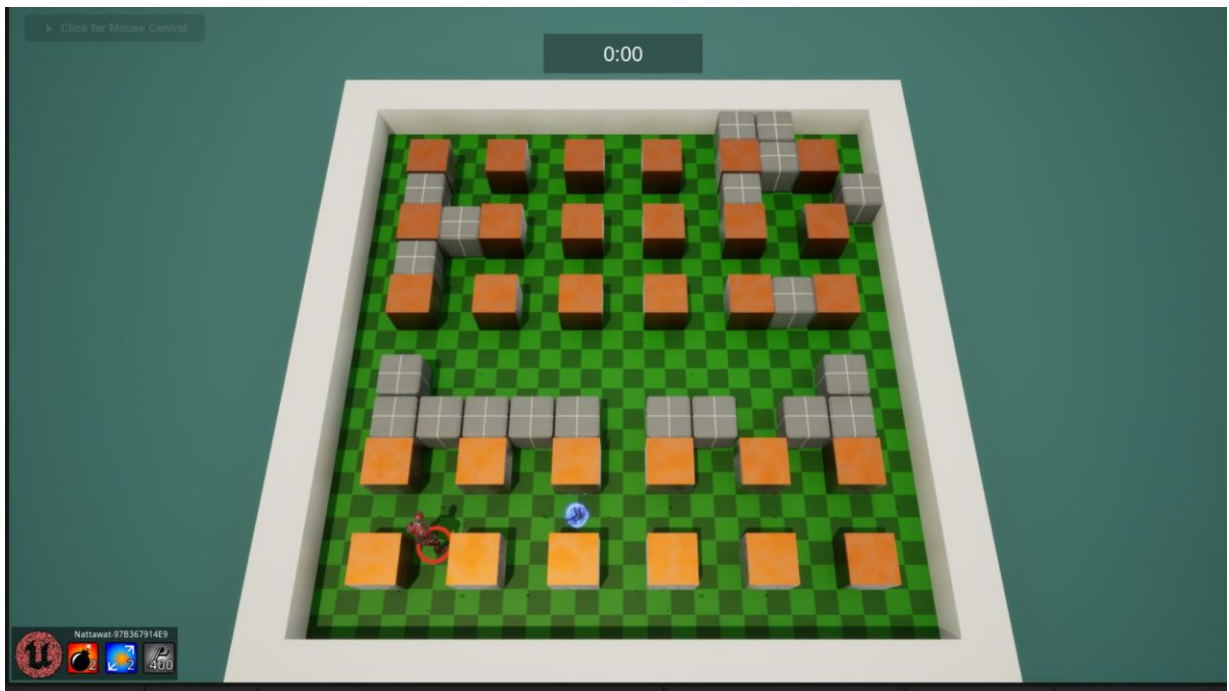


Figure 38: Example gameplay of "3D Bomber Man Game"

3D Shooting Game



Figure 39: Example gameplay of "3D Shooting Game"

ScF 03_01 (Eng)



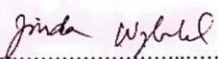
The Human Research Ethics Committee of Thammasat University (Science), (HREC-TUSc) Room No 110, Piyachart Building, 1st Floor, Thammasat University Rangsit Campus, Prathumthani 12121 Thailand, Tel: 0-2564-4440 ext.7358 E-mail: ecscu3@tu.ac.th

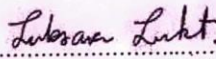
COA No. 071/2566

Certificate of Approval

Project No. : 66AH074
Title of Project : Application of Inertial Measurement Unit (IMU) Sensors for Measuring Kinematic of Wheelchair Compared with 2D Motion Analysis Program
Principle Investigator : Sairag Saadprai
Place of Proposed Study/Institution: Faculty of Allied Health Sciences, Thammasat University

The Human Research Ethics Committee of Thammasat University (Science), Thailand, has approved the above study project in accordance with the compliance to the Declaration of Helsinki, the Belmont report, CIOMS guidelines and the International practice (ICH-GCP).

Signature: 
 (Assoc. Prof. Jinda Wangboonskul, Ph.D.)
 Chairman of the Human Research Ethics
 Committee of Thammasat University (Science).

Signature: 
 (Assoc. Prof. Laksana Laokiat, Ph.D.)
 Secretary of the Human Research Ethics
 Committee of Thammasat University (Science).

Date of Approval: August 24, 2023 **Approval Expire date:** August 23, 2024
Progressing Report Due: July 24, 2024

The approval documents including

- 1) Research proposal Version 2/ 17.07.2023
- 2) Principal Investigator's Curriculum Vitae Version 2/ 17.07.2023
- 3) Patient/Participant Information Sheet Version 3/11.08.2023
- 4) Informed Consent Form Version 2/ 17.07.2023
- 5) Flyer Version 2/ 17.07.2023
- 6) Physical Activity Readiness Questionnaire Plus (PAR-Q) 2019) Version 2/ 17.07.2023
- 7) Case record form Version 2/ 17.07.2023

Figure 40: shows the Certificate of Approval Application of Inertial Measurement Unit (IMU) Sensors for Measuring Kinematic of Wheelchair Compared to 2D Motion Analysis Program



**MEMORANDUM OF UNDERSTANDING
BETWEEN
CENTER OF EXCELLENCE IN CREATIVE ENGINEERING DESIGN AND DEVELOPMENT,
THAMMASAT UNIVERSITY
AND
MATSUNAGA MANUFACTORY CO., LTD.**

In order to encourage closer academic ties, Center of Excellence in Creative Engineering Design and Development, Thammasat University and Matsunaga Manufactory Co.,Ltd. enter into an agreement of cooperation to establish programs of exchange and collaboration in areas of interest and benefit to both institutions. This agreement will serve as a general framework for cooperation between the two institutions and is intended to facilitate discussions of more specific programs of collaboration. In particular, the two institutions agree to promote the following collaborations:

1. Coordination of such activities as joint research and lectures;
2. Exchange of data, documentation, and research materials in fields of mutual interest;
3. Other mutually beneficial projects.

A designated person/committee at each institution shall oversee and facilitate implementation of this agreement in cooperation with other appropriate administrators at the respective institutions.

The scope of activities under this agreement shall be determined by the funds regularly available at both institutions for the types of collaboration undertaken and by such other financial assistance as may be obtained by either institution.

Unless required by law, an announcement, circular or other public disclosure including promotional materials such as newsletters, brochures, flyers or annual reports, referring to the contents or subject matter of this MOU, must not be made or permitted by a party without the prior written approval of the other party.

The present Memorandum comes into effect from the date of its signature by THE PARTIES and will remain valid for the period of 3 (three) years. Either institution wishing to terminate this agreement shall notify the other institution in writing at least 60 (sixty) days prior to termination. Such notification shall not affect the status of students who have already been accepted for programs initiated prior to the termination of this agreement. Any decision to renew or renegotiate the terms of this agreement shall be made on the basis of consultations between the two institutions.

For: Center of Excellence in Creative
Engineering Design and Development

Asst.Prof.Bunyong Rungruangdouyboon, Ph.D.
Head of Center of Excellence in Creative
Engineering Design and Development

Asst.Prof.Sairag Saadprai, Ph.D.
Member of Center of Excellence in Creative
Engineering Design and Development

Date: 7 March, 2024

For: Matsunaga Manufactory Co.,Ltd.

Mr.Shinji Hayashi
Managing Director Matsunaga (Thailand) Co.,Ltd

Miss Bussaya Jampangam
Businesses Development Matsunaga (Thailand) Co.,Ltd

Date: 7 March, 2024

Figure 41: Memorandum of Understanding between Center of Excellence in Creative Engineering Design and Development, Thammasat University with Matsunaga Manufactory which is one of the largest wheelchair companies in Japan

Has your Project/Device been patented? If yes, please provide details.

Thailand Patent/Petty Patent	Thailand Identity
1.Movement and exercise analysis equipment	Thailand Petty Patent no. 21587
2.Wheelchair motion analysis system	Thailand Petty patent application no. 2303002152
3.Exercise equipment for wheelchair users with training programs and fitness tracking systems.	Thailand Patent application no.2401001717
4. Physical Motion Tracking Sensor Controlled Wheelchair	Thailand Patent application no. 2401002438

เลขที่อนุสิทธิบัตร 21587



อสป/200 - ข

อนุสิทธิบัตร

อาศัยอำนาจตามความในพระราชบัญญัติสิทธิบัตร พ.ศ. 2522
ซึ่งแก้ไขเพิ่มเติมโดยพระราชบัญญัติสิทธิบัตร (ฉบับที่ 3) พ.ศ. 2542
อธิบดีกรมทรัพย์สินทางปัญญาออกอนุสิทธิบัตรฉบับนี้ให้แก่

มหาวิทยาลัยธรรมศาสตร์

สำหรับการประดิษฐ์ตามรายละเอียดการประดิษฐ์ ข้อถือสิทธิ และรูปเขียน (ถ้ามี) ดังที่ปรากฏในอนุสิทธิบัตรนี้

เลขที่คำขอ 2203000781

วันขอรับอนุสิทธิบัตร 30 มีนาคม 2565

ผู้ประดิษฐ์ นางสาวรัก สอาดไพร และ นายศุภชัย วรพจน์พิศุทธิ์

ชื่อที่แสดงถึงการประดิษฐ์ อุปกรณ์วิเคราะห์การเคลื่อนไหวและการออกกำลังกาย

21587

ให้ผู้ทรงอนุสิทธิบัตรนี้มีสิทธิและหน้าที่ตามกฎหมายว่าด้วยสิทธิบัตรทุกประการ



ออกให้ ณ วันที่ 11 เดือน พฤษภาคม พ.ศ. 2566

หมดอายุ ณ วันที่ 29 เดือน มีนาคม พ.ศ. 2571



รองอธิบดีกรมทรัพย์สินทางปัญญา ปฏิบัติราชการแทน
อธิบดีกรมทรัพย์สินทางปัญญา
ผู้ออกอนุสิทธิบัตร

พนักงานเจ้าหน้าที่



Ref. 256601037283090

- หมายเหตุ
1. ผู้ทรงอนุสิทธิบัตรต้องชำระค่าธรรมเนียมรายปีเริ่มตั้งแต่ปีที่ 5 ของอายุอนุสิทธิบัตร มิฉะนั้น อนุสิทธิบัตรนี้จะสิ้นสุดอายุ
 2. ผู้ทรงอนุสิทธิบัตรจะขอชำระค่าธรรมเนียมรายปีล่วงหน้าโดยชำระทั้งหมดในคราวเดียวได้
 3. ภายใน 90 วันก่อนวันสิ้นสุดอายุอนุสิทธิบัตร ผู้ทรงอนุสิทธิบัตรมีสิทธิขอต่ออายุอนุสิทธิบัตรได้ 2 ครั้ง มีกำหนดคราวละ 2 ปี
 4. การอนุญาตให้ใช้สิทธิตามอนุสิทธิบัตรและการโอนอนุสิทธิบัตรต้องทำเป็นหนังสือและจดทะเบียนต่อพนักงานเจ้าหน้าที่

010089

Figure 42: Petty patent of Movement and exercise analysis equipment, No. 21587

6. Abstract

Please write a short summary of the overall Project/Device and should not exceed 200 words.

The "AI Motion Tracking System for Monitoring Health and Activity through Innovative Motion-Controlled Gaming for Wheelchair Users" addresses the challenges wheelchair users face in everyday life. We aim to improve physical health, reduce anxiety and depression risks, and enhance the overall quality of life for wheelchair users using advanced sensors, AI to predict motions, a movement tracking system, a vital sign monitor, and a fall detection system to provide real-time data for themselves and caregivers. Our project was inspired by the smartwatch and the Nintendo Wii, which emphasizes wheelchair users to do physical activities in daily life and exercise through our developed games while monitoring health and activities in real time. Additionally, we will integrate the wheelchairs with smart home systems, for controlling devices such as doors, lights, and air conditioners and further develop the system to enhance convenience and support independent living.